

Essential Insights: Module 7



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What is Agentic AI in Business Terms

Sustaining Innovation in AI

- Sustaining innovation enhances existing processes without changing organisational structures
- In AI, this improves capabilities like content generation and data analysis within existing tools

Defining Agentic AI

- Agentic AI builds on Generative AI to enable autonomous multi-step workflows across functions
- It can plan, decide, and act independently within predefined organisational boundaries

From Theory to Business Application

Shift in Focus

- Moves from understanding how agentic systems reason to how they operate within enterprise environments
- Emphasis on embedding AI into workflows to deliver tangible operational improvements

Workflow Transformation

- Enhances processes such as procurement and finance through automation and decision-making
- Enables faster execution and improved coordination across tasks

Traditional vs Agentic AI

Traditional Digital Systems

- AI acts as a support tool by identifying issues or assisting with communication
- Human managers remain responsible for decision-making and execution

Agentic AI Systems

- Independently identifies problems, evaluates options, and executes solutions
- Operates proactively without waiting for human input

Real-World Example – Procurement

Traditional Scenario

- AI highlights disruptions such as supply chain issues
- Procurement manager manually identifies alternatives and adjusts operations

Agentic Scenario

- System autonomously identifies disruption, selects suppliers, and updates production plans
- Executes end-to-end resolution with minimal human intervention

Ensuring Process Continuity

Challenges in Modern Operations

- Workflows often stall due to disruptions, exceptions, or incomplete information
- Manual intervention slows down resolution and increases inefficiencies

Role of Agentic AI

- Navigates uncertainty and ensures workflows continue despite disruptions
- Maintains consistent progress through adaptive decision-making

Real-World Application – Ocado

Automated Fulfilment Systems

- Agentic systems coordinate thousands of robots in real time within warehouses
- Continuously adjust routing and inventory handling based on live demand

Business Impact

- Reduces executional friction and improves operational efficiency
- Enables human workers to focus on higher-value decision-making

Evolution of Automation

Task Automation

- Focuses on completing single repetitive tasks such as data extraction
- Lacks awareness of broader workflows and requires human continuation

Workflow Automation

- Connects tasks into structured sequences for improved efficiency
- Limited flexibility due to reliance on predefined, rule-based logic

Agentic AI Transformation

Goal-Oriented Execution

- Moves beyond rigid workflows to systems driven by high-level objectives
- Adapts dynamically to changing inputs and business contexts

Advanced Capabilities

- Analyses customer or operational context to personalise actions
- Determines optimal actions such as communication, timing, and engagement strategy

Impact on Human Work

Goal-Oriented Execution

- Eliminates manual coordination required to connect tasks and workflows
- Reduces inefficiencies caused by fragmented processes

Strategic Focus

- Frees managers from supervising processes and handling exceptions
- Enables focus on higher-value activities such as strategy and decision-making

Human-in-the-Loop Design

Importance of Oversight

- Full autonomy is not always suitable in high-risk or regulated environments
- Certain decisions must remain under human control

Allocation of Responsibilities

- AI handles data collection, analysis, and preparation tasks
- Humans retain authority over final approvals and critical decisions

Balancing Speed and Risk

Trade-Off Consideration

- Increased autonomy improves speed but may introduce higher risk
- Greater human oversight enhances control but reduces efficiency

Decision Frameworks

- Organisations must assess workflows based on risk and reversibility of errors
- Escalation protocols define when human intervention is required

Practical Implementation at Scale

Real-World Example – Amazon

- Agentic systems manage routine operations such as inventory routing and scheduling
- Low-risk decisions are handled autonomously to improve efficiency

Governance in Practice

- High-impact decisions such as contracts or disruptions require human review
- Ensures balance between automation and organisational control

Key Takeaways

Core Insights

- Agentic AI represents a shift from task support to workflow ownership
- Enables continuous, autonomous execution across business functions

Strategic Importance

- Enhances efficiency, scalability, and operational resilience
- Positions organisations for future transformation and competitive advantage